

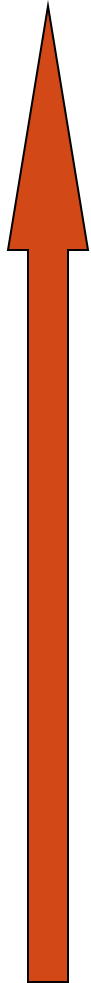
OSI

- Open Systems Interconnection (OSI) is a set of internationally recognized, non-proprietary standards for networking and for operating system involved in networking functions

OSI Model



PDU's And the OSI Model



Layer	PDU Name
7.) Application	Data
6.) Presentation	Data
5.) Session	Data
4.) Transport	Segment
3.) Network	Packet
2.) Data Link	Frame
1.) Physical	Bits



Application

Presentation

Session

Transportation

Network

Data Link

Physical

T
O
P

B
O
T
T
O
M

Application Layer: Network Applications



HTTP HTTPS FTP
NFS FMTP DHCP
SNMP TELNET
POP3 IRC NNTP



File Transfer



FTP



Web Surfing



HTTP/S



Emails



SMTP

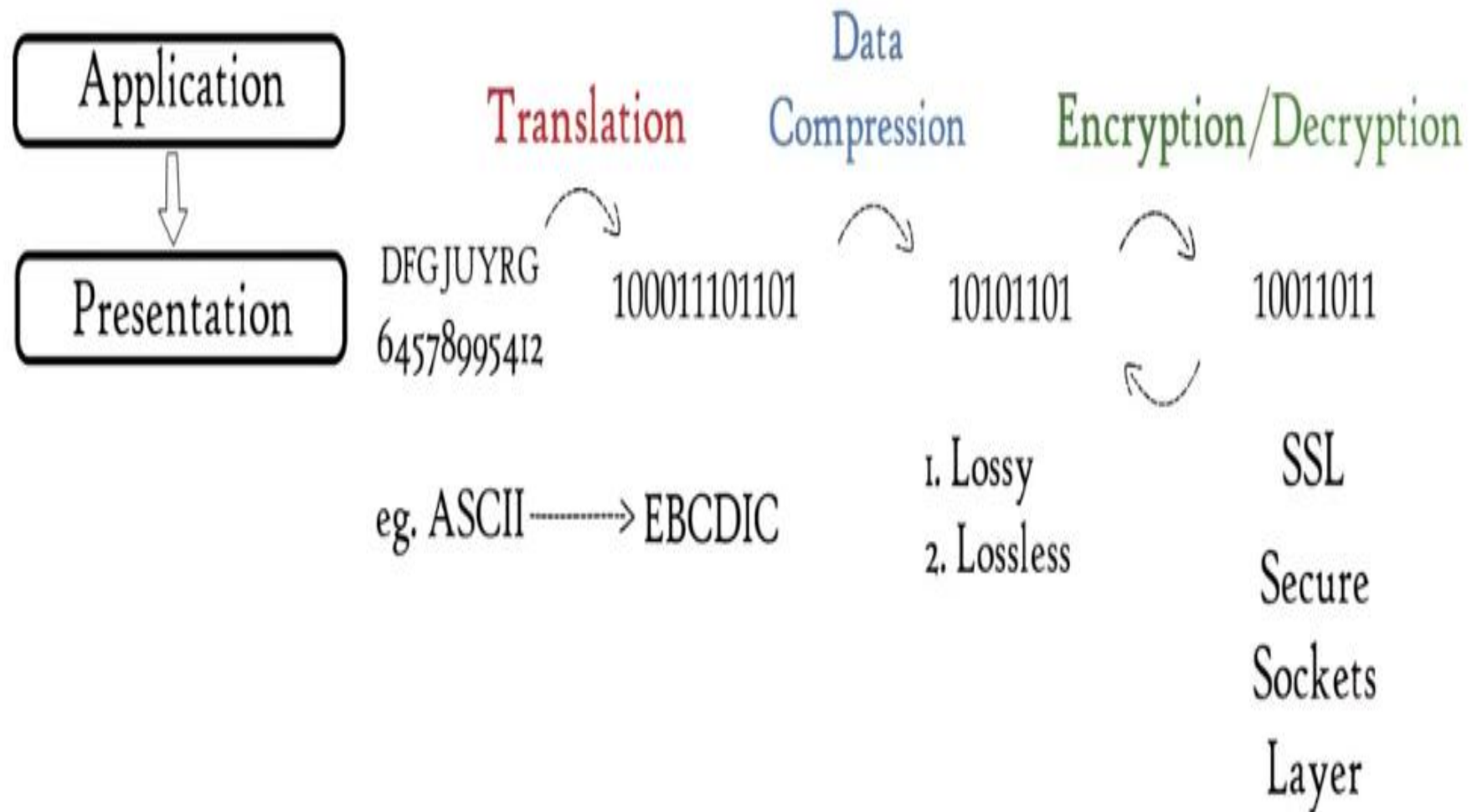
Virtual
Terminals

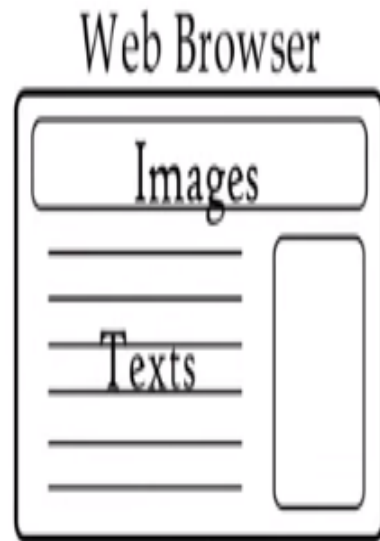
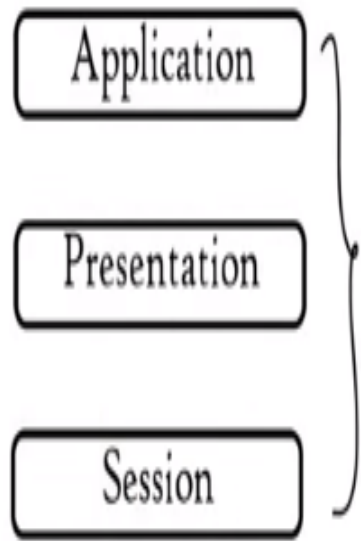


Telnet

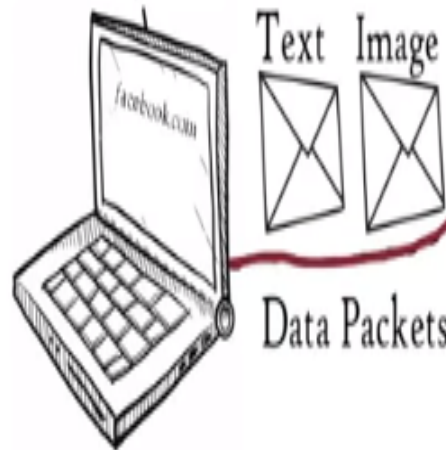
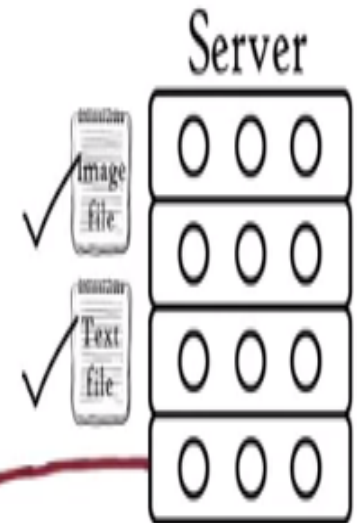


Presentation Layer





Session Layer



Authentication ✓

Authorization ✓

Session Management

Transport Layer

Segmentation

Flow Control

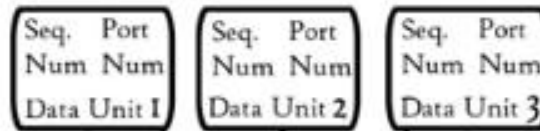
Error Control

Connection and
Connectionless Tx

Segmentation:

Data

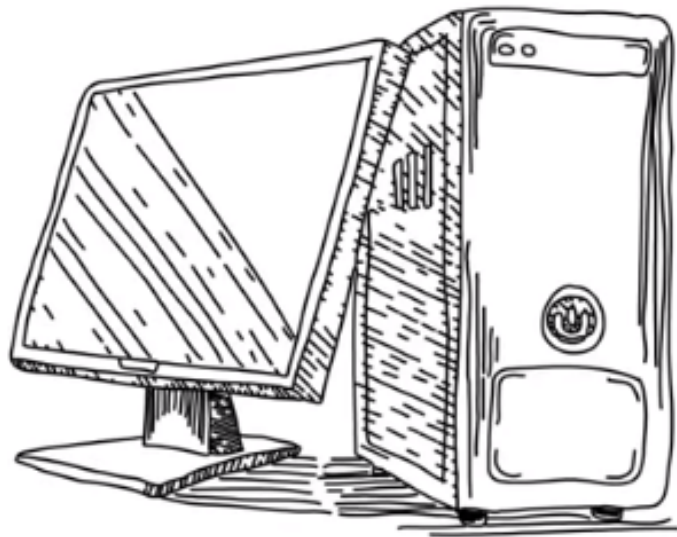
Segments



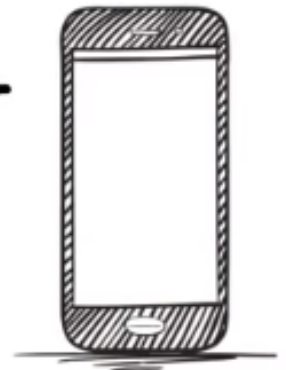
Data

Flow Control:

XXXX Mbps



100 Mbps

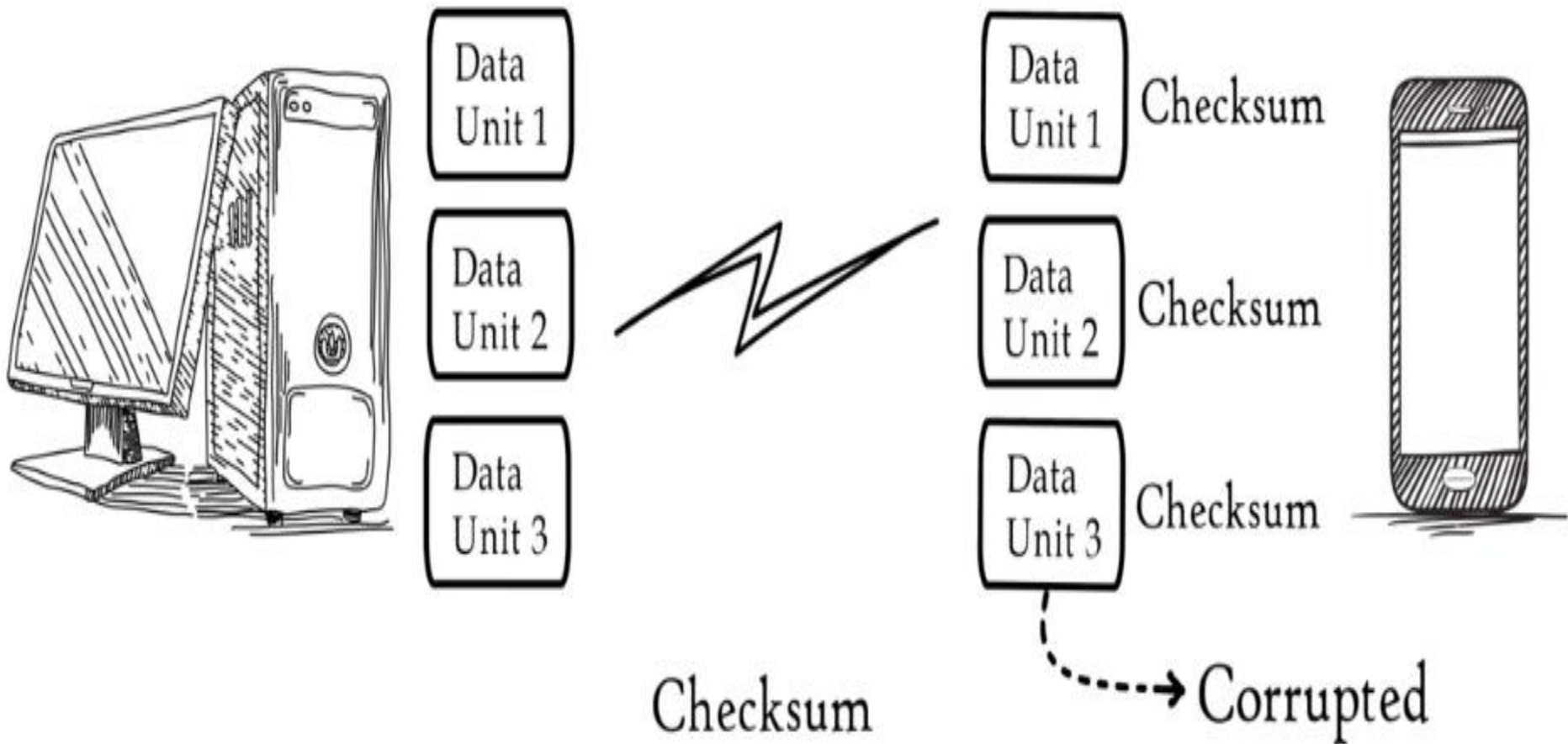


10 Mbps

Error Control:



Automatic Repeat Request



Transport Layer

Services:

- Connection-oriented Transmission
- Connectionless Transmission

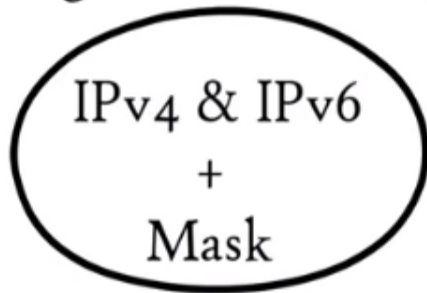
Protocols:

- > Transmission Control Protocol (TCP)
- > User Datagram Protocol (UDP)

Network layer



Logical Addressing

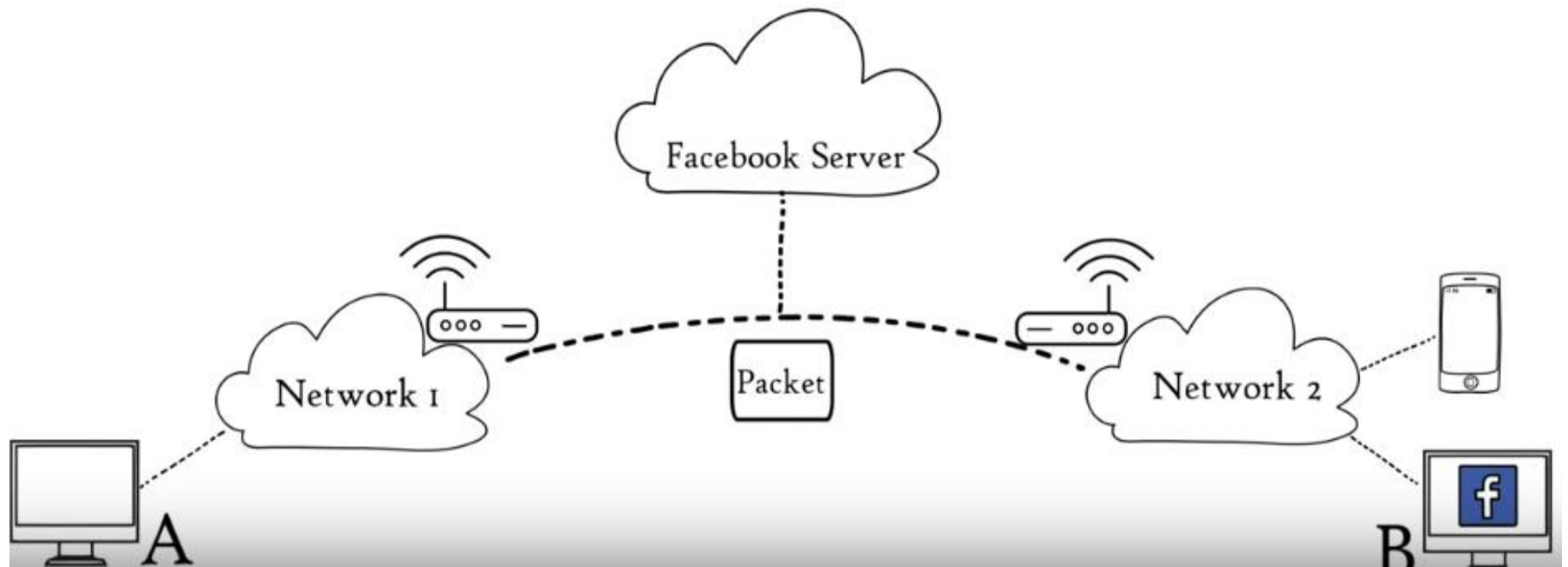


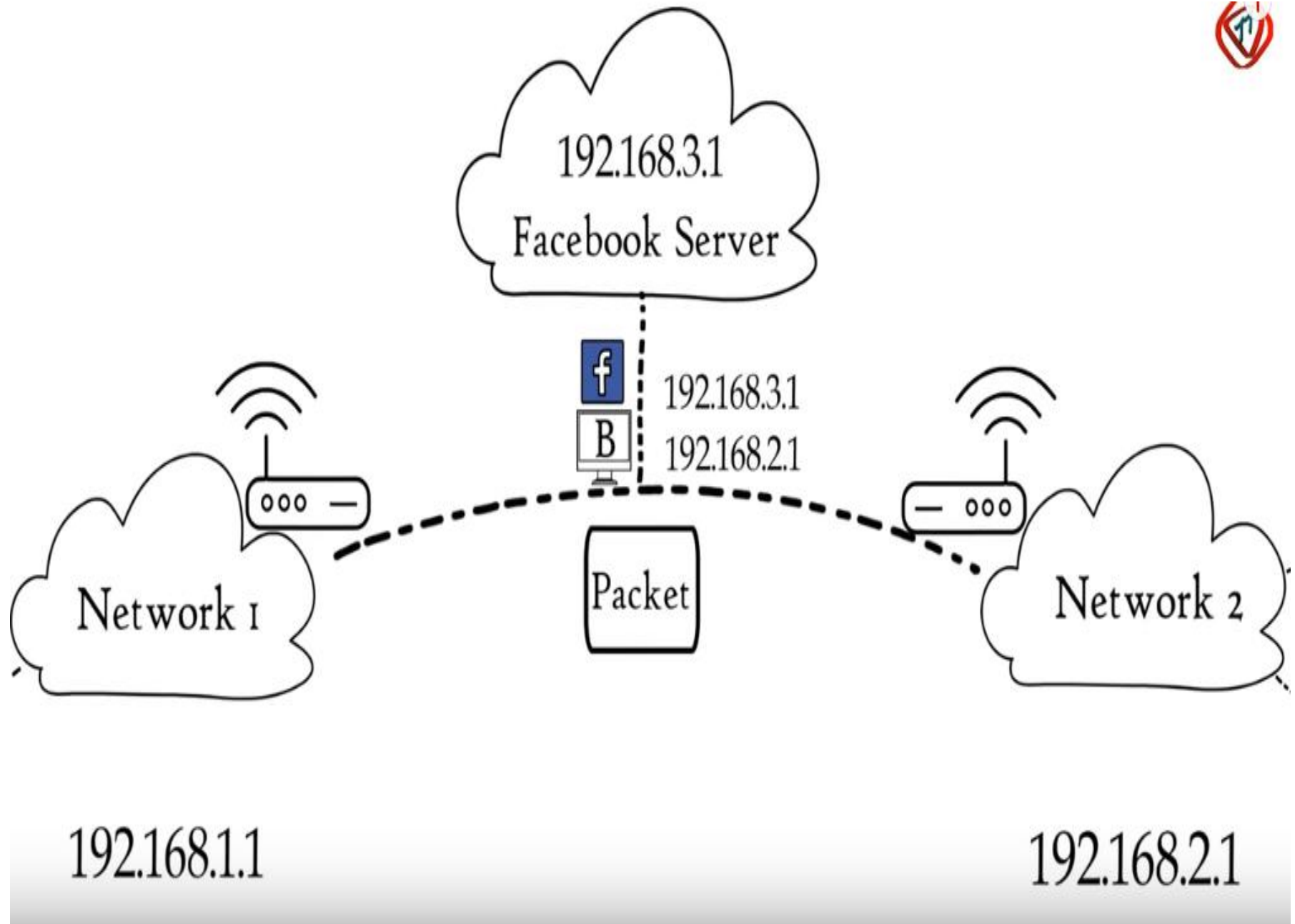
Path determination

Routing



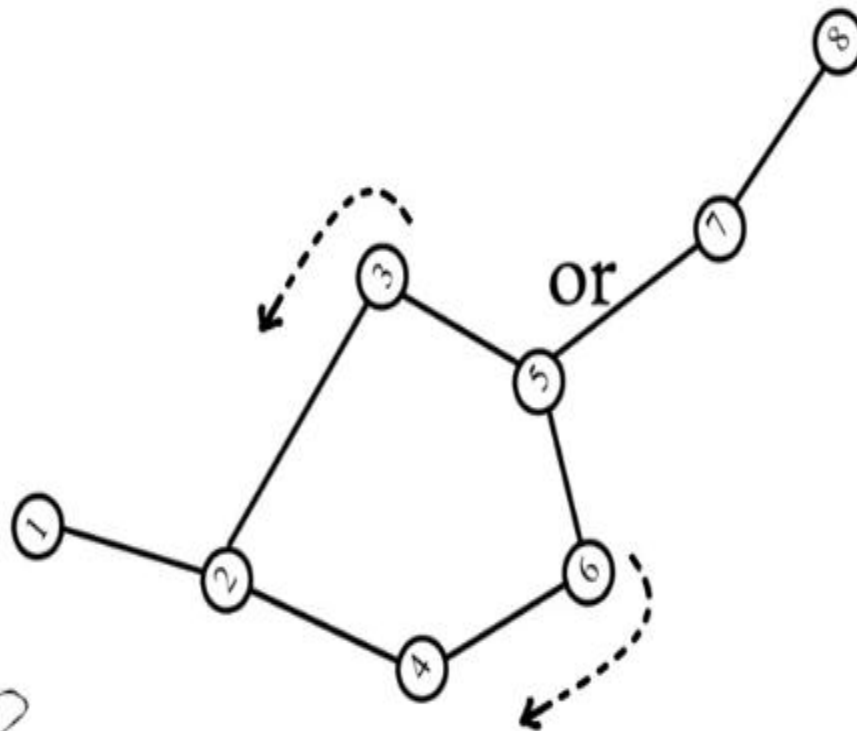
Routing





Path determination

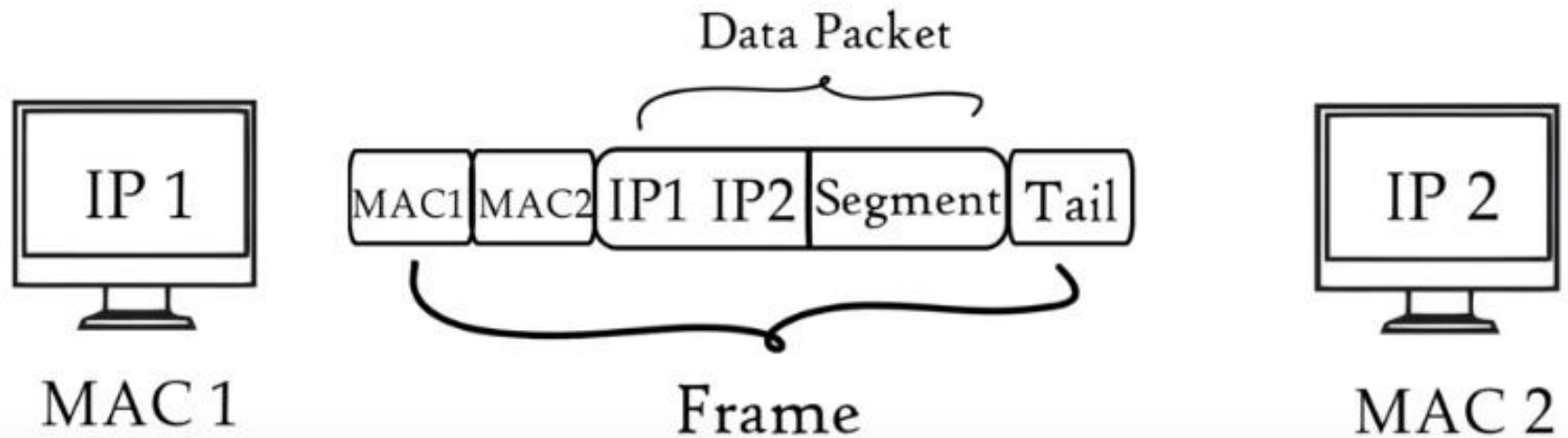
↑
OSPF
BGP
IS-IS

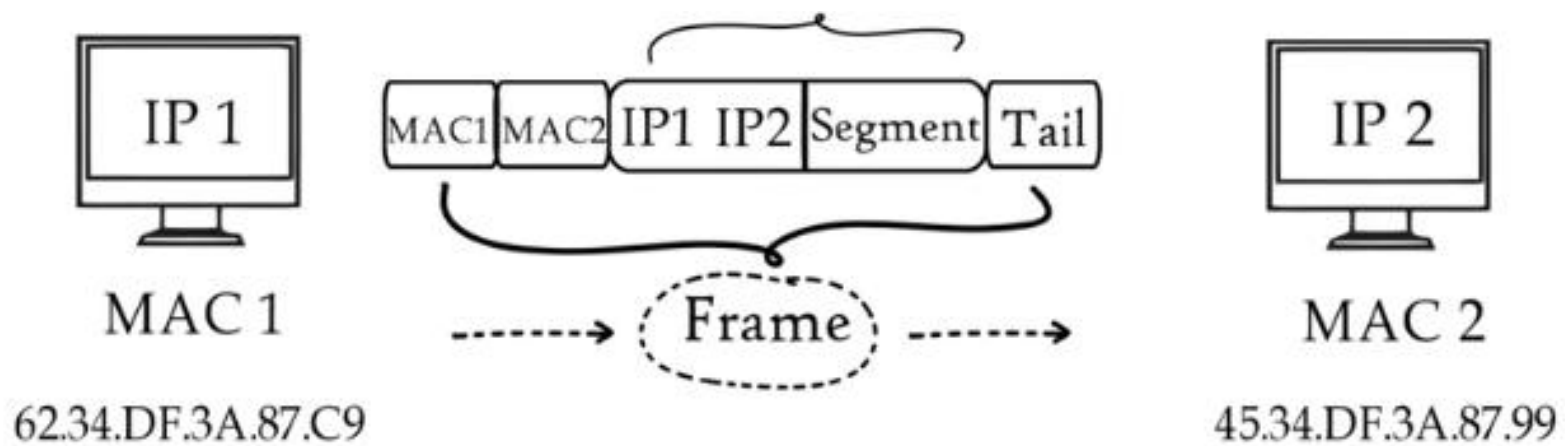


Sender

Data Link Layer

- Logical addressing : Network layer
- Physical addressing : Data Link layer





Controls how data is placed
and received from the media



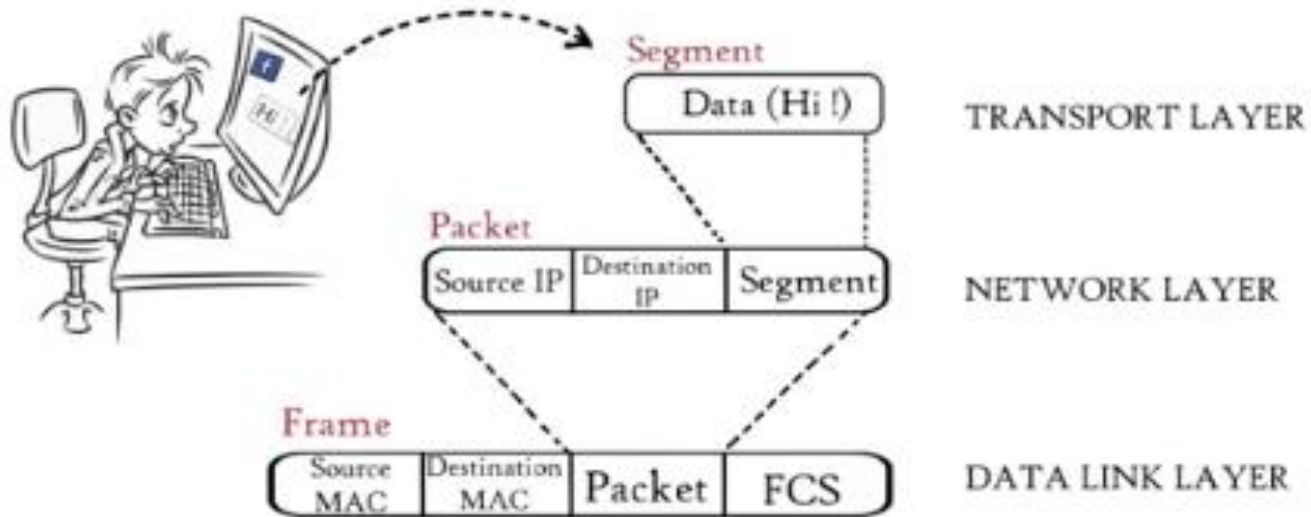
(Media Access Control)
(Error Detection)

DATA LINK LAYER



(LAN Cable, Optical Fiber, Air)

Physical Layer

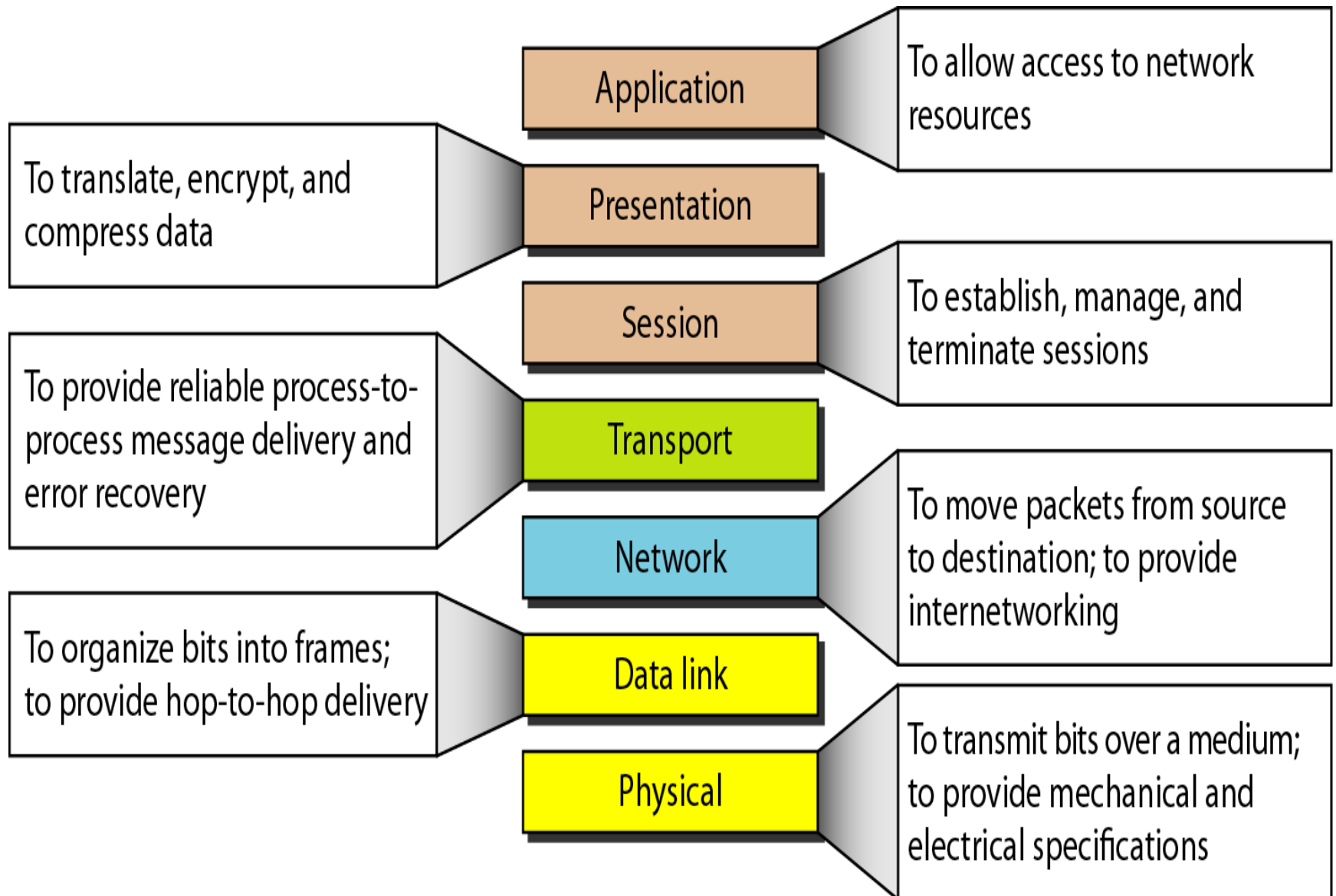


1001011101101100101110110111011

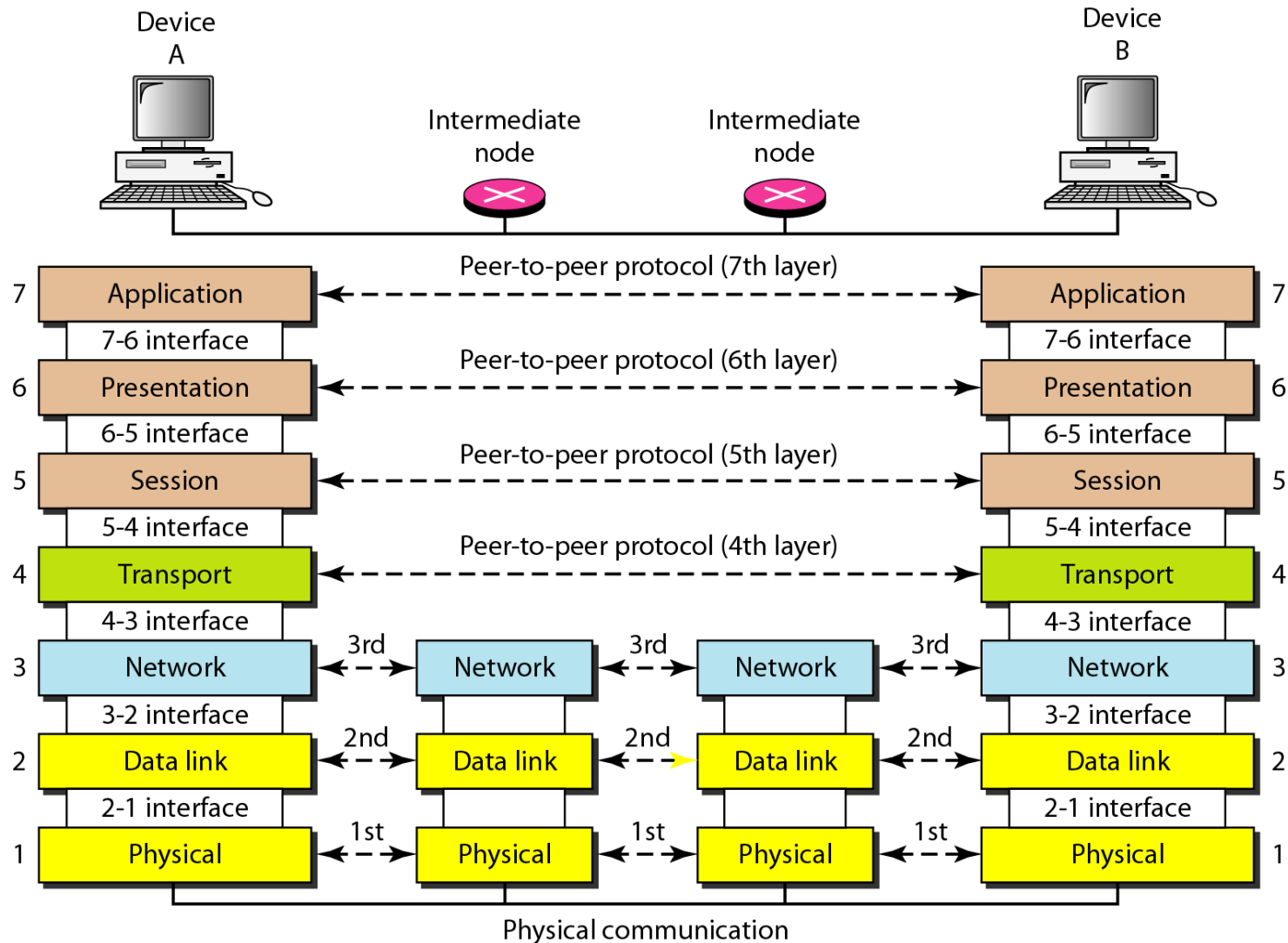
BITs



Summary



The interaction between layers in the OSI model



An exchange using the OSI model

